

Notes: Tobin (Econometrica, 1958)

Estimation of Relationships for Limited Dependent Variables

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Limited dependent variables	2
2.2	Latent index and observed outcome	2
2.3	Empirical variables in the illustration	2
3	Models and estimation specifications	3
3.1	Distributional assumptions	3
3.2	Probability and density	3
3.3	Conditional expectation	3
3.4	Maximum likelihood and inference	3
4	Statistical identification and empirical design	3
5	Tables and figures	4
5.1	Figure 1: Engel curve with a lower limit; Figure 2: $\lambda_{min}(x) = -Z(x)/Q(x)$	4
5.2	Table I: Sums of Squares and Cross Products; Table II: Iterative Estimation of Parameters	4
5.3	Table III: Iterative Estimation Assuming $\beta_2 = 0$; Table IV: Estimated Variances and Covariances	5
5.4	Table V: Calculation of Expected Values; Figure 3a: $X_2 = 0$; Figure 3b: $X_2 = 2$; Table VI: Observed Values	5
6	Short summary	6
7	Conclusion	7
7.1	Core conclusion	7
7.2	Economic intuition	7
7.3	Why OLS and probit are incomplete	7
7.4	Lasting contribution	7
7.5	Practical takeaway	7
7.6	Limitations and extensions	7

1 Big picture

- **Research question.** How should researchers estimate relationships when the dependent variable is censored or limited, with many observations piled up at a boundary but continuous values away from that boundary?
- **Core example.** Durable-goods expenditures are zero for many households, but positive buyers spend widely varying amounts.
- **Main contribution.** Tobin proposes a maximum-likelihood estimator that combines the participation margin and the expenditure margin. This became the foundational Tobit model.
- **Why OLS fails.** OLS treats zeros as ordinary realizations and imposes a linear observed conditional mean.
- **Why probit alone fails.** Probit models the probability of being at the limit but ignores the size of positive outcomes.
- **Economic takeaway.** Extensive and intensive margins must be estimated jointly for limited dependent variables.

2 Data and measurement

2.1 Limited dependent variables

The outcome W has a lower limit L . A substantial share of observations is exactly at that limit, while non-limit observations take continuous values above the limit. The limit may vary across observations, although the illustration uses $L = 0$.

Interpretation: The observed distribution has a discrete mass at the boundary and a continuous part away from the boundary.

2.2 Latent index and observed outcome

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_m X_{mi}.$$
$$W_i = L_i \quad \text{if } Y_i - \varepsilon_i < L_i, \quad W_i = Y_i - \varepsilon_i \quad \text{if } Y_i - \varepsilon_i \geq L_i.$$

Interpretation: The latent index is linear, but the observed outcome is censored.

2.3 Empirical variables in the illustration

The dependent variable is durable-goods expenditure divided by disposable income. X_1 is the age category of the household head, and X_2 is liquid assets divided by disposable income. There are 183 limit observations and 552 non-limit observations.

Interpretation: The example is not a causal design; it illustrates how censored outcomes should be estimated.

3 Models and estimation specifications

3.1 Distributional assumptions

$$\varepsilon_i \sim N(0, \sigma^2), \quad P(x) = \Phi(x), \quad Q(x) = 1 - \Phi(x), \quad Z(x) = \phi(x).$$

3.2 Probability and density

$$\Pr(W_i = L_i \mid Y_i, L_i) = Q\left(\frac{Y_i - L_i}{\sigma}\right),$$

$$f(W_i \mid W_i > L_i, Y_i, L_i) = \frac{1}{\sigma} Z\left(\frac{Y_i - W_i}{\sigma}\right).$$

Interpretation: Limit observations enter as probabilities; positive observations enter as densities.

3.3 Conditional expectation

Let $k_i = (Y_i - L_i)/\sigma$. Then

$$E(W_i \mid Y_i, L_i) = L_i Q(k_i) + Y_i P(k_i) + \sigma Z(k_i).$$

Interpretation: The expected observed outcome is nonlinear even though the latent index is linear.

3.4 Maximum likelihood and inference

Using $a = 1/\sigma$ and $I_i = a_0 + a_1 X_{1i} + \dots + a_m X_{mi}$,

$$\log L = \sum_{limit} \log Q(I_i - aL_i) + \sum_{nonlimit} [\log a + \log Z(I_j - aW_j)].$$

$$LR = -2(\log L_R - \log L_U) \sim \chi^2.$$

Interpretation: Tobin solves the nonlinear likelihood equations using Newton iteration and uses likelihood-ratio tests for restrictions.

4 Statistical identification and empirical design

The model identifies the latent linear relationship and the error variance by combining information from the boundary mass and the continuous positive outcomes. The key assumptions are that the censoring rule is known, the limit is observed, errors are normal and homoskedastic, and regressors are exogenous.

Interpretation: The paper is a methodological contribution. Its identification is statistical rather than causal.

5 Tables and figures

5.1 Figure 1: Engel curve with a lower limit; Figure 2: $\lambda_{\min}(x) = -Z(x)/Q(x)$

Read: Figure 1 motivates the censored-outcome problem. Figure 2 plots the inverse-Mills-ratio term used in the likelihood conditions and starting-value approximation.

Economic message: The figures connect economic corner outcomes to nonlinear likelihood estimation.

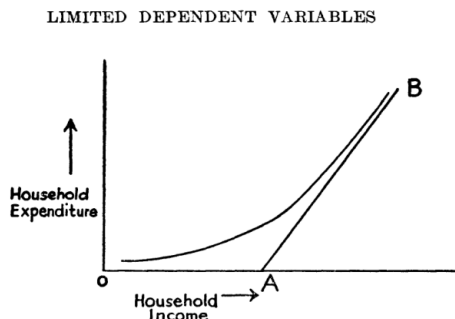
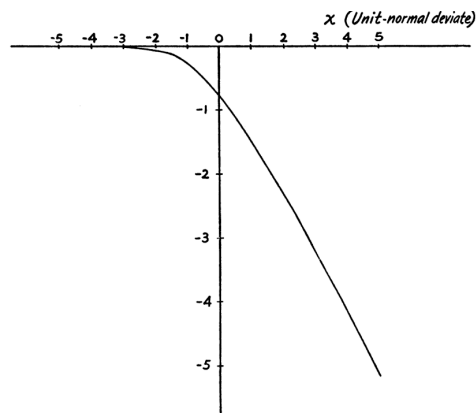


FIGURE 1

ific example, many—indeed, most—households would reject on automobiles or major household durable goods during a

(a) Figure 1: Engel curve with a lower limit.



(b) Figure 2: inverse-Mills-ratio function used for starting values.

Figure 1 shows why the observed conditional mean bends toward zero. Figure 2 shows the nonlinear adjustment term generated by censored observations.

5.2 Table I: Sums of Squares and Cross Products; Table II: Iterative Estimation of Parameters

Read: Table I separates limit and non-limit observations. Table II reports unrestricted maximum-likelihood iterations and final estimates.

Economic message: Zeros and positives are used together but enter the likelihood differently.

65 or more years 6

TABLE I
SUMS OF SQUARES AND CROSS PRODUCTS

183 limit observations					552 non-limit observations				
	$X_0 = 1$	X_1	X_2	W		$X_0 = 1$	X_1	X_2	W
$X_0' = 1$	183				$X_0 = 1$	552			
X_1'	824	4056			X_1	1976	8060		
X_2'	102.15	552.03	402.3333		X_2	168.06	751.54	255.6740	
W'	0	0	0	0	W	61.449	207.598	20.559	13.113087

(a) Table I: sums of squares and cross products.

TABLE II
ITERATIVE ESTIMATION OF PARAMETERS

	a_0	a_1	a_2	a
Initial trial values	1.326	-.2200	.0330	7.984
First derivatives	-4.398	-21.759	-1.812	.898
Second derivatives a_0	-680.557			
a_1	-2542.416	-10,805.486		
a_2	-238.41	-1128.653	-535.223	
a	61.449	207.598	20.559	-21.772
Indicated changes	.0152	-.00507	.00199	.0376
Second trial values	1.3407	-.2251	.0350	8.022
First derivatives	-.047	.292	.064	.002
Second derivatives a_0	-680.260			
a_1	-2540.666	-10,795.522		
a_2	-238.26	-1,127.900	-535.893	
a	61.449	207.598	20.559	-21.688
Indicated changes	-.0015	.00037	.00001	-.00064
Final estimates	1.3392	-.2247	.0350	8.022
Standard errors	(.118)	(.0295)	(.0495)	(.252)

(b) Table II: unrestricted iterative maximum-likelihood estimation.

Table I documents the split between boundary and non-boundary observations. Table II shows the numerical mechanics of Tobit estimation.

5.3 Table III: Iterative Estimation Assuming $\beta_2 = 0$; Table IV: Estimated Variances and Covariances

Read: Table III imposes the restriction that liquid assets have no effect. Table IV reports variance-covariance matrices.

Economic message: Likelihood-based inference evaluates restrictions in the censored regression model.

TABLE III
ITERATIVE ESTIMATION OF PARAMETERS ASSUMING THAT $\beta_2 = 0$

	a_0	a_1	a
Initial trial values	1.337	-.219	8.040
First derivatives	-2.841	-16.124	-.001
Second derivatives: a_0	-680.419		
a_1	-2541.428	10,799.12	
a	61.449	207.598	-21.652
Indicated changes	.010	-.004	-.010
Second trial values	1.347	-.223	8.030
First derivatives	-.456	-2.017	+.116
Second derivatives: a_0	-680.179		
a_1	-2539.988	-10,790.472	
a	61.449	207.598	-21.674
Indicated changes	.001	.0003	-.005
Final estimates	1.347	-.223	8.030
Standard errors	(.117)	(.028)	(.252)

(a) Table III: restricted estimation assuming $\beta_2 = 0$.

In the approximation used to obtain initial trial values, the function $-Z(x)/Q(x)$ was approximated by the tangent at the point $x_0 = .67$, so that $Q(x_0) = .25$,

TABLE IV
ESTIMATED VARIANCES AND COVARIANCES OF PARAMETER ESTIMATES

	a_0	a_1	a_2	a
a_0	+.0139			
a_1	-.00318	+.000867		
a_2	+.000880	-.000454	+.00245	
a	+.00987	-.00115	+.000470	+.0635
On assumption that $\beta_2 = 0$:				
	a_0	a_1	a	
a_0	+.0136			
a_1	-.00302	+.000784		
a	+.00970	-.00106	+.0635	

(b) Table IV: estimated variances and covariances.

The restricted estimates are close to the full estimates, suggesting the liquid-asset coefficient is statistically weak in the illustration.

5.4 Table V: Calculation of Expected Values; Figure 3a: $X_2 = 0$; Figure 3b: $X_2 = 2$; Table VI: Observed Values

Read: Table V translates latent-index estimates into probabilities and expected observed values. Figures 3a and 3b compare the Tobit expected-value curve with the OLS regression line. Table VI reports observed values by age group.

Economic message: Tobit estimates a linear latent relationship but implies a nonlinear observed conditional expectation.

34

JAMES TOBIN

TABLE V
CALCULATION OF EXPECTED VALUES

X_1	$X_2 = 0$				$X_2 = 2$			
	$I = 1.3392 - .2247X_1$	Calculated probability of buying $P(I)$	$Z(I)$	Calculated expected value $E(W) = (IP + Z)/8.022$	$I = 1.4092 - .2247X_1$	Calculated probability of buying $P(I)$	$Z(I)$	Calculated expected value $E(W) = (IP + Z)/8.022$
0	1.3392	.910	.163	.172	1.4092	.921	.148	.180
1	1.1145	.867	.214	.147	1.1845	.882	.197	.155
2	.8898	.813	.267	.123	.9598	.832	.252	.131
3	.6651	.747	.319	.102	.7351	.768	.304	.108
4	.4404	.670	.362	.082	.5104	.695	.350	.088
5	.2157	.585	.390	.064	.2857	.612	.383	.070
6	-.0090	.497	.399	.049	.0610	.524	.398	.054
7	-.2337	.408	.388	.037	-.1637	.435	.393	.040
8	-.4584	.323	.359	.026	-.3884	.350	.370	.029

(a) Table V: calculated probabilities and expected values.

TABLE VI
OBSERVED VALUES
(all values of X_2)

X_1	Proportion buying	Observed average value of W
1	.812	.124
2	.884	.118
3	.862	.098
4	.751	.083
5	.664	.145
6	.516	.047

(b) Table VI: observed values.

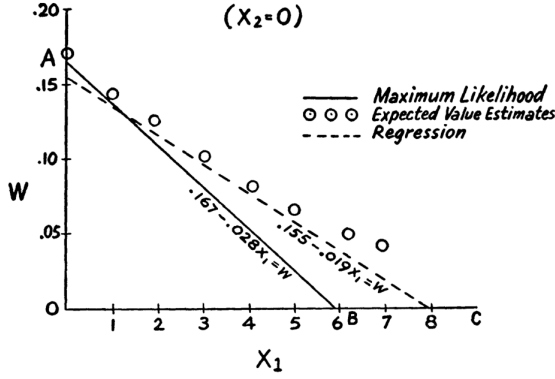


FIGURE 3a

(a) Figure 3a: estimated relationship for $X_2 = 0$.

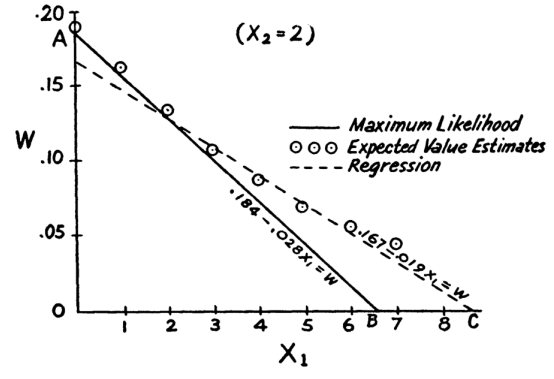


FIGURE 3b

(b) Figure 3b: estimated relationship for $X_2 = 2$.

Table V shows both the probability of positive expenditure and expected spending. Figures 3a and 3b show why OLS can be misleading when the conditional mean naturally bends toward the censoring point.

6 Short summary

- The paper studies dependent variables with many observations at a known lower or upper limit.
- The likelihood combines a boundary probability and a continuous normal density.
- The conditional mean of the observed variable is nonlinear.
- The durable-goods example illustrates the method with zero expenditures and positive spending amounts.
- The paper introduced the foundation of the Tobit estimator.

7 Conclusion

7.1 Core conclusion

Tobin shows that limited dependent variables require a model that accounts for both censoring at the boundary and continuous variation away from the boundary.

7.2 Economic intuition

Observed zeros are not ordinary regression residuals. They are censored realizations that reveal that the latent desired value was below the feasible limit.

7.3 Why OLS and probit are incomplete

OLS ignores the probability mass at the limit, while probit ignores the positive outcome amount. Tobit uses both.

7.4 Lasting contribution

The paper created the template for censored regression and corner-solution models in applied econometrics.

7.5 Practical takeaway

Researchers should distinguish true censoring from hurdle choices, sample selection, and measurement censoring before choosing a limited-dependent-variable model.

7.6 Limitations and extensions

The original model assumes normal homoskedastic errors, exogenous regressors, and known censoring limits. Later work relaxes these assumptions through heteroskedastic, semiparametric, selection, and endogenous-regressor models.